FUTURE DRILLING COSTS SIMULATION

TEAM 4

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Overview

Compagnie Pétrolière et Gazière, Inc. (hereafter referred to as "Company") faces a big risk in drilling operations due to the possibility of drilling in dry holes, resulting in temporal and monetary losses. The Company has requested a simulation of project-level well outcomes along with statistics on the project's financial success for Phase 2 of their exploration process.

To address these requests, our team simulated well outcomes and calculated statistics for the project's Net Present Value (NPV), Value at Risk (VaR), and Conditional Value at Risk (CVaR). After considering oil price projections for the next 15 years along with the Company's provided revenue and risk estimates, our simulations guaranteed non-zero profit. We therefore encourage the Company to pursue this investment.

Methodology

The following section describes the structure and preparation of the data for our simulation.

Data Used

Our team utilized a dataset containing estimated drilling costs for crude oil, natural gas, and dry wells from 1991-2007. The dataset included geometric annual changes on these costs. We also incorporated oil price projections from 2026 to 2050 for revenue calculations.

Methods

Using provided distributional and production assumptions for a single well, we ran a Monte Carlo simulation on the probability of the number of producing wells in the entire project. The number of wells in each project was modeled as a Uniform distribution between 10 and 30, and each well's production outcome was simulated using a Binomial distribution with an 85% probability of producing and 15% probability of being a dry well.

We used this information, combined with our Phase 1 estimates of 2025 drilling costs, to perform a Monte Carlo simulation on the distribution of NPV for the entire project. Our analysis assumes that wells that produce will do so for 15 years and uses a 10% weighted average cost of capital (WACC) in the NPV calculation. We calculated expected risk and return from dry wells and producing wells.

In addition to the Monte Carlo simulation, we developed a base-case NPV model using mean or most likely values for all inputs. This deterministic model provides a benchmark estimate of project value and allows for comparison with the Monte Carlo simulation of NPV, which captures uncertainty and risk not captured by the base-case model.

Analysis

This section presents our simulation of project-level well production and NPV.

Simulating Well Production

Before simulating well production, the Company informed us that the probability of a producing well was 85%. They also specified that the number of planned wells in the project is expected to be uniformly distributed between 10 and 30. With this information, we ran a simulation of the number of wells in the project, including dry wells. Figure 1 visualizes the distribution of the proportion of producing wells.

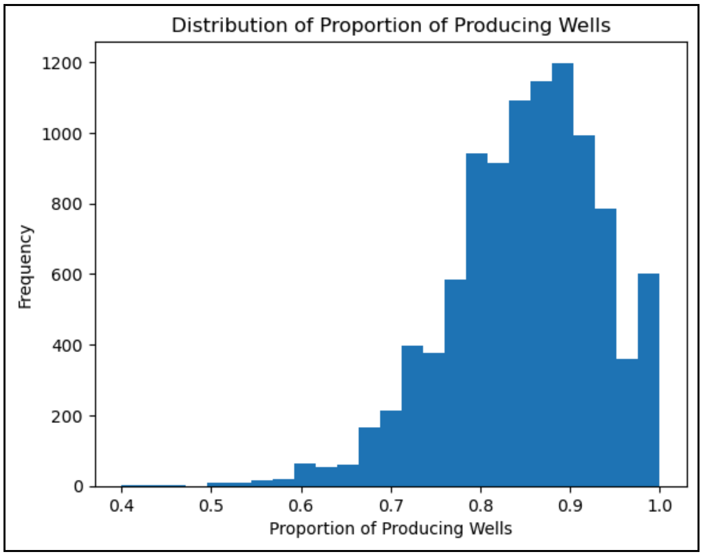

Figure 1: Distribution of proportion of producing wells.

The high concentration of values near 0.85 along with the left skew in Figure 1 indicates that in general, we anticipate the proportion of producing wells to be between 70% and 95%. While there is still a moderate chance of the proportion falling below 70%, the chance of this proportion following below 50% is nearly zero. Therefore, the Company can expect around 85% of the project's planned wells to produce.

Simulating Project NPV

We leveraged our analysis from Phase 1 to simulate the project's NPV. During Phase 1, we concluded that drilling costs per well are projected to increase to an average value of approximately $4.4M in 2025, when assuming a Normal distribution for the simulation. We also found that the probability of costs falling below the initial estimate is approximately 7.85%. After reporting this information, the Company provided additional metrics related to anticipated project costs and revenue to simulate the project's NPV. Table 1 summarises the expenses that occur at Year 0 before oil production begins.

Table 1: Year 0 Expenses.

Expense	Description	Cost	Relevant Distribution
Drilling Costs	Simulated drilling costs calculated during Phase 1.	$4.4M per well *(expected)*	Varies by year; see Phase 1 report for more details.
Drill Rights	The right to drill on land surrounding a project well.	$960 per acre *(per well)*	Total acres per well ~ $Normal(600,\ 50^2)$
Seismic Data	Data purchased to choose the optimum well location.	$43k per section *(per well)*	Total sections per well ~ $Normal(3,\ 0.35^2)$
Completion Cost	Cost of oil production at optimum sustainable rate *(given that the well is not dry).*	Varies by well	Cost per well ~ $Normal(\$390k,\ \$50k^2)$
Overhead	Annual costs of project team salary and benefits, based on total time spent on well.	Varies by well	Cost per well: ● Most likely: $215k ● Minimum: $172k ● Maximum: $279.5k

Table 1 indicates that pre-production costs often vary by well, which emphasizes the importance of accurately estimating how many producing wells will be included in the project. Fortunately, these values are only accrued during Year 0, with the exception of annual overhead costs. Overall, pre-production costs amount to approximately $5.7M per producing well.

This project must generate enough overall revenue to cover these major expenses with an acceptable profit margin. To help simulate expected revenue, the Company provided information on production risk factors, which are summarized in Table 2.

Table 2: Production risk factors.

Factor	Description	Distribution/Equation
Initial Production Rate (IP)	Rate that oil is produced at Year 1, measured in barrels of oil per day (BOPD).	IP's ~ $LogNormal(420,\ 120^2)$ Underlying distribution ~ $Normal(6,\ 0.28^2)$
Decline Rate (DR)	Decline in production rate as reservoir energy and volumes deplete over time, measured in BOPD.	DR ~ $Uniform(15,\ 32)$
Annual Production Rate (APR)	Overall production rate per year, measured in BOPD.	$Rate_{YearEnd} = (1 - DR) \times Rate_{YearBegin}$
Annual Production Volume (APV)	Overall production volume per year, measured in BOPD.	$Oil_{VolumeYear} = 365 \times \frac{Rate_{YearBegin} + Rate_{YearEnd}}{2}$
IP/DR Correlation	Correlation coefficient between IP and DR, imposed by the Company.	$Corr(IP, DR) = 0.64$

From Table 2, our team noticed that the correlation coefficient between the IP and DR is moderately strong. This value therefore indicates that the decline rate increases as the initial production rate increases. Additionally, every well has its own rate of decline, so the distributions in Table 2 were used in our simulation to account for this uncertainty.

Next, the Company shared its estimates of certain revenue risk factors, as described in Table 3.

Table 3: Revenue risk factors.

Factor	Description	Distribution/Equation
Annual Oil Price	Projected oil prices per barrel for the next 15 years, provided by the U.S. Energy Information Administration.	Price ranges over time: • Most likely: $85.26 to $97.68 • Minimum: $45.75 to $49.74 • Maximum: $166.26 to $185.28
Net Revenue Interest (NRI)	Percentage of oil revenue retained by the Company after royalties. Constant over lifetime per well.	NRI ~ $Normal(75\%,\ 2\%^2)$
Annual Revenue	Pre-royalty revenue generated in a given year.	$Revenue_{Year} = OilPrice_{Year} \times APV_{Year}$

Table 3 includes the royalties that must be paid out on each well, with each well potentially having different royalty values. This project should therefore include enough wells with adequate production volume to ensure that post-royalty revenue meets an acceptable threshold.

Finally, the Company included operating costs and taxes for West Texas, which are specified in Table 4.

Table 4: Operating expenses.

Factor	Description	Distribution/Equation
Production Costs	Cost of manpower and hardware involved in the production process, measured in dollar amount per barrel.	Cost per barrel $\sim Normal(\$2.25,\ \$0.30^2)$
Severance Taxes	Constant state taxes levied on produced oil and gas, applied after the NRI.	4.6% of annual revenue.

Table 4 shows that production costs and severance taxes make up the Company's operating expenses. These expenses are subtracted from the Company's gross sales to obtain their net sales.

These four tables summarize the collective risks and revenue streams that impact the NPV of the entire project. With this information in mind, we simulated the distribution of NPV for the entire project.

Results & Recommendations

This section outlines the results of our analysis and gives recommendations for future decision-making.

Results

The outcome of our team's base-case NPV was \$331M, which provides the benchmark estimate of project value. This value aligns with the mean NPV from the Monte Carlo simulations, which was \$332.6M. Both of these numbers indicate that the project will be profitable for the Company.

Figure 2 depicts the entire distribution of simulated NPV values, which incorporates our simulations of the risk and revenue stream factors. The navy line represents the CVaR; the red line represents the VaR.

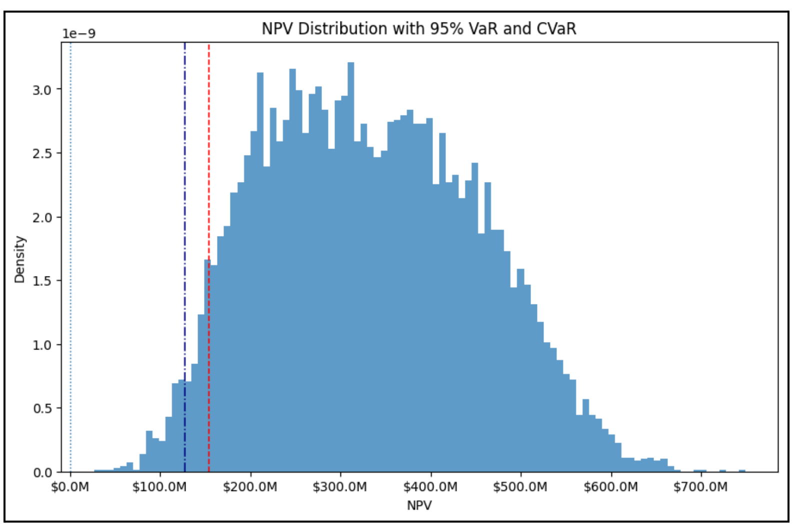

Figure 2: Simulated distribution of the entire project's NPV.

The distribution in Figure 2 immediately indicates that the project is guaranteed to yield a positive NPV for the Company, with 0% of the area falling on or below the $0 threshold. The red VaR line indicates that in the worst 5% of project outcomes, project NPV is estimated to be $182.4M or lower. Similarly, the navy CVaR line indicates that within these worst 5% of outcomes, the expected project NPV is estimated to be $147.9M. For further comparisons across possible project outcomes, Table 5 provides risk and shortfall metrics at three different confidence levels. Note that the confidence levels correspond with a tail probability of $1 - Confidence\%$.

Table 5: VaR and CVaR statistics at different confidence levels.

Confidence	VaR (5% Tail Threshold)	NPV* - VaR Loss	CVaR (Average 5% Tail NPV)	NPV* - CVaR Loss
90%	$182.4M	$150.3M	$147.9M	$184.8M
95%	$154.2M	$178.4M	$127.0M	$205.6M
99%	$109.8M	$222.8M	$90.0M	$242.6M

**NPV = $332.6M*

Table 2 indicates that across all three confidence intervals, the CVaR estimate is at least $19M lower than the VaR threshold. This difference indicates a strong tail risk, since project NPV is expected to be at least $19M lower than the VaR threshold in the worst 5% of possible project outcomes. Conversely, when looking at the difference between expected NPV and VaR/CVaR, Table 2 indicates that the Company is expected to lose $178.4M or more in the worst 5% of outcomes. In these scenarios, the average loss is expected to be about $205.6M, which is $27.2M more than the worst 5% NPV threshold.

Fortunately, none of the scenarios are expected to result in a negative NPV. Even in the worst 1% of project outcomes, project NPV is projected to be $90M on average. While the Company has not specified a target NPV, our team strongly believes that this value indicates a relatively safe project investment.

Recommendations

Assuming that the historical data provided by the Company continues to apply to present scenarios *(i.e., barring any major climate event or policy change)*, our model predicts a 100% probability of positive NPV. Therefore, our team recommends proceeding with this investment. The CVaR indicates that even in the worst 5% of scenarios, the average loss is only $205.6M lower than the mean, which still leaves the company with $127M in profit. Meanwhile, in the most likely scenarios, this project would offer the Company approximately $332.6M in profits.

Conclusion

Our team performed a simulation of well outcomes for the Company's prospective drilling operation. We used historical data on estimated drilling costs, Company-provided estimates for risk and revenue, and additional outside assumptions *(such as the WACC)* to calculate project NPV in different scenarios. Our model found a 100% probability of profit on the investment, with the worst case scenario producing an average NPV of $90M. In the most likely scenarios, our team expects the Company to earn $332.6M on average; for this reason, we highly recommend that the Company move forward with this project.